Soumya Tyagi

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Summary

AI/ML Engineer with hands-on experience creating machine learning solutions, LLM-based systems, retrieval-augmented generation (RAG) pipelines, and agentic workflows using Python. Experienced in developing retrieval pipelines, integrating LLMs with external tools, and building AI applications using LangChain, LangGraph, ChromaDB, FastAPI, and Streamlit. Strong foundation in data structures, algorithms, and core computer science concepts.

Education

Vellore Institute of Technology September 2023 - May 2027
B.Tech in Computer Science and Engineering (Artificial Intelligence and Machine Learning)
CGPA: **8.43/10**

Technical Skills

Programming: Python, SQL, Java (DSA)
AI & Machine Learning: Machine Learning, Deep Learning, NLP, Transformers, Scikit-learn, TensorFlow, Hugging Face
Generative AI & Agentic AI: LLMs, Embeddings, RAG, AI Agents, Tool Calling, LangChain, LangGraph, MCP, ChromaDB, FAISS
AI Engineering & Operations: FastAPI, Streamlit, Model Evaluation, RAG Evaluation, LLMOps, AI Security
Tools & Core CS: Git, GitHub, Docker, AWS, Google Colab, OOP, DBMS, Computer Networks

Experience

Videonza Technologies (OPC) Private Limited May 2026 - Jul 2026
AI/ML Development Intern | Hybrid

- Improved model accuracy from **82% to 91%** through data preprocessing, feature engineering, and hyperparameter tuning using Scikit-learn and PyTorch.
- Performed exploratory data analysis and feature engineering on structured datasets to identify data quality issues and prepare features for model training.
- Evaluated model performance using appropriate validation metrics and optimized training workflows to support reliable model selection and deployment.

Projects

RepoMind AI — Agentic RAG GitHub Repository Intelligence | *Python · LangChain · ChromaDB · Sentence Transformers · Groq LLaMA · GitHub API* Aug 2026

- Engineered an **AI-driven GitHub codebase assistant** using LangChain, ChromaDB, and Groq, developing 4+ specialized tools for multi-step repository search and code analysis.
- Improved RAG retrieval accuracy by 25% over the baseline pipeline by integrating query expansion, multi-query retrieval, cross-encoder re-ranking, and context compression.
- Expanded code search coverage by 3× for complex queries by replacing single-step retrieval with agentic tool routing across files, languages, and repo structures.

AI Production Incident Investigator | *Python · LangGraph · LangChain · Groq · RAG · ChromaDB · FastAPI · Streamlit* Oct 2026

- Engineered an **AI-based Production Incident Investigator** using Python, LangChain, LangGraph, RAG, and LLMs to analyze production incidents through logs, system documentation, and historical incident data.
- Designed a 6-stage LangGraph workflow covering incident analysis, hypothesis generation, evidence retrieval, evaluation, and re-investigation, using ChromaDB and Sentence Transformers for semantic search.
- Built FastAPI and a Streamlit interface that generate structured incident reports with root cause, confidence scores, supporting/conflicting evidence, and actionable remediation steps.

Achievements

Johns Hopkins Health Hackathon

- Contributed to a nutrition recommendation system by creating a responsive frontend and integrating backend APIs for food analysis and personalized dietary insights.

Inter-College Basketball Tournament

- Secured **1st Place** with the college basketball team, demonstrating teamwork, discipline, leadership, and the ability to perform under competitive pressure.

Certifications

- AWS Certified AI Practitioner - AWS
- Applied Machine Learning in Python - Coursera